



The role of information and communication technologies-based extension in agriculture: application, opportunities and challenges

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The Role of ICTs in Agricultural Extension

Information and Communication Technologies (ICTs) have emerged as transformative tools in agricultural extension, offering innovative solutions to bridge the information gap that has long hindered agricultural productivity, especially among smallholder farmers in developing countries. The article presents a systematic review of 49 empirical studies, analyzing the impact of ICT-based interventions-such as mobile apps, SMS messaging, educational videos, and voice calling services-on awareness, adoption of agricultural practices, crop yields and farmer incomes. The review highlights both the significant opportunities and persistent challenges associated with the adoption and effectiveness of ICTs in agricultural extension.

Importance of ICTs in Agricultural Extension

Access to timely and relevant information is a critical determinant of agricultural productivity. Traditionally, farmers have relied on research institutions, extension agents and non-governmental organizations for information about new technologies. However, the conventional extension system faces several limitations, including a shortage of human resources, lack of accountability, and the delivery of generalized messages that may not be relevant to specific local contexts. Moreover, face-to-face extension campaigns often struggle to reach remote areas and are further constrained during crises, such as the COVID-19 pandemic, which limit physical contact.

The rise of ICTs, including the Internet and mobile phones, has led to the emergence of digital advisory services, commonly referred to as e-extension or ICT-based extension. These platforms provide farmers with timely and site-specific advice on agricultural practices, input use, crop planting dates, local weather predictions, market prices, and access to social networks. The growing availability of ICT applications has resulted in a significant expansion of digital extension services, with observational and experimental studies evaluating their impacts on outcomes such as knowledge, awareness, adoption of technologies, choice of market channels, health, profit and income.

Awareness and Knowledge: The impact of ICT interventions on awareness and knowledge of agricultural technologies is mixed. While some studies found that ICTs increased farmers' knowledge of new technologies, others reported no significant impact. The effectiveness of ICTs in raising awareness depends on the design of the messaging, the format of the information delivery, and the level of user engagement. Passive information delivery, such as radio broadcasts or basic SMS messages, is often insufficient to change farmers' knowledge or practices. In contrast, interactive formats, such as educational videos and mobile apps, have been more successful in increasing awareness and knowledge retention.

Adoption of Agricultural Technologies: ICTs have been particularly effective in promoting the adoption of agricultural technologies. SMS messaging, mobile apps, educational videos, and voice calls have all been shown to increase the adoption of good agricultural practices, such as improved nutrient management, integrated pest management, and the use of improved seeds. The accessibility and interactivity of these platforms enable timely reminders and personalized guidance, which are crucial for supporting farmer decision-making. However, the adoption of more complex technologies, such as new seed varieties, remains challenging and often requires additional training and support.

Crop Yields: The majority of studies (74%) reported that ICT-based interventions led to higher crop yields. The most common technologies that positively impacted yields were smartphone apps, SMS, voice calls, and videos. These platforms provide farmers with timely and relevant information on best practices, such as optimal planting dates, fertilizer application, and pest management. However, the impact of ICTs on yields is not universal and depends on the quality of the information provided, the level of user engagement, and the availability of complementary inputs and services.

Farmer Incomes: ICT interventions have also been shown to increase farmer incomes by providing timely production and market information. Mobile phones, SMS and voice calls connect farmers to market prices and buyers, enabling them to make informed decisions about when to sell their crops and how to negotiate better prices. However, the impact on income is not always significant, as standalone information often fails to alter outcomes. For example, access to market information alone may not be sufficient to increase incomes if farmers lack access to credit, inputs, or infrastructure to implement recommended practices

Opportunities and Challenges

This article highlights several opportunities and challenges associated with the use of ICTs in agricultural extension. One of the main opportunities is the ability of ICTs to reach a large number of farmers, including those in remote and underserved areas. The versatility and interactivity of mobile apps and educational videos make them particularly effective in providing personalized and hyperlocal advice. Additionally, ICTs can facilitate two-way communication, enabling farmers to provide feedback and ask questions, which can enhance the relevance and effectiveness of the information provided.

However, several challenges remain. Limited digital literacy and access to smartphones are significant barriers to the reach of ICT-based interventions, particularly among smallholder farmers. The success of ICT interventions also depends on the quality of the messaging, the level of user engagement, and the availability of complementary services, such as training and access to inputs. Moreover, the impact of ICTs on agricultural outcomes is often context-specific, and interventions that are successful in one setting may not be effective in another.

Recommendations for Effective ICT-Based Extension

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Conclusion

ICTs have the potential to significantly advance agricultural development outcomes by providing timely and relevant information to farmers. However, their success depends on well-designed messaging, adequate user training, and integration with complementary interventions. The review underscores the importance of customizing ICT interventions to local contexts and capacities, and highlights the need for a comprehensive approach that addresses the multiple constraints facing smallholder farmers. By leveraging the strengths of different ICT platforms and adopting participatory and context-specific approaches,

agricultural extension services can more effectively support the adoption of good agricultural practices, increase crop yields and enhance farmer incomes.

References

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